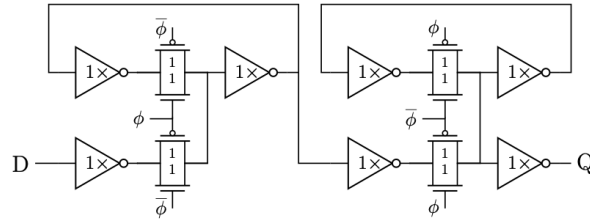


EE5311 LAB Tutorial-5 Report

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Problem Statement:

Using a $1\times$ inverter($W_n/L_n = 0.42\mu\text{m}/0.15\mu\text{m}$ and $W_p/L_p = 0.84\mu\text{m}/0.15\mu\text{m}$), draw the below schematic of a D flip-flop in **xschem**.



Measure the setup time(t_{setup}) and clock-Q delay (t_{CQ}) for the above flip-flop **driving a $1\times$ inverter** for the following transitions at Q:

- (a) rising
- (b) falling

Assume that the clock ϕ and its complement $\bar{\phi}$ are available as inputs. Use the following waveform for:

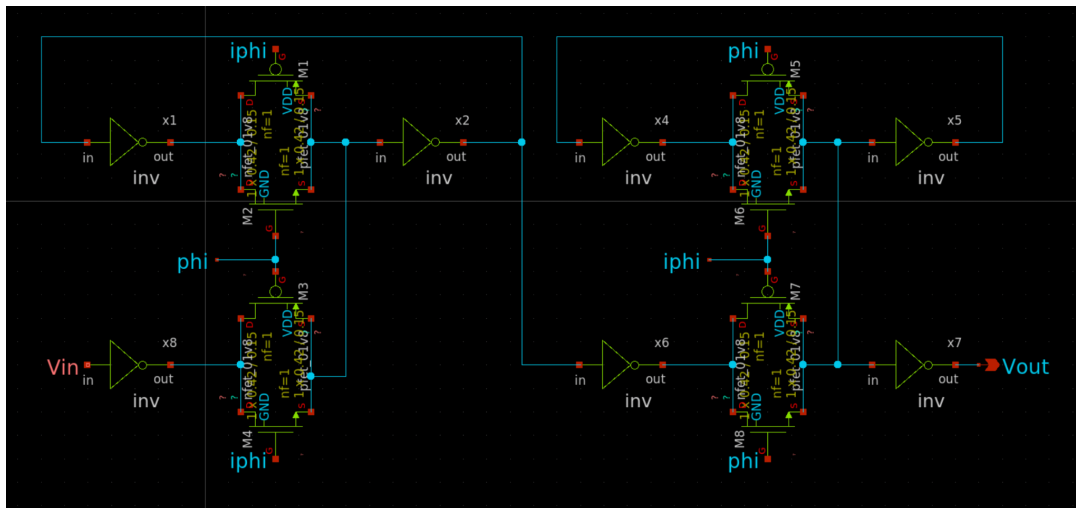
ϕ : pulse 0 1.8 10n 5p 5p 9.99n 20n and

$\bar{\phi}$: pulse 1.8 0 10n 5p 5p 9.99n 20n.

Assume 5ps rise and fall times for the D input. Sweep the transition time of D and find the time at which t_{DQ} is minimum. t_{DC} and t_{CQ} for this transition time gives the setup time and clock-Q delay.

Question 1.A (Rising Edge):

Schematic :-



Schematic

```

s1
.param td = 0.5n
.ic V(Vout) = 0
.ic V(Vd) = 0
.control
let Nsim = 40
let tdcvec = vector(Nsim)
let tcqvec = vector(Nsim)
let tdqvec = vector(Nsim)
let tdvvec = vector(Nsim)
let index = 0
while index<Nsim
    let tdv = 9.8n + (index*((0.12n)/Nsim))
    alterparam td = $&tdv
    reset
    tran 2p 50n
    meas tran tdc trig v(Vd) val = 0.9 rise=1 targ v(phi) val=0.9 rise=1
    meas tran tcq trig v(phi) val = 0.9 rise=1 targ v(Vout) val=0.9 rise=1
    let tdcvec[index] = $&tdc
    let tcqvec[index] = $&tcq
    let tdqvec[index] = $&tdc + $&tcq
    let tdvvec[index] = $&tdv
    let index = index+1
end
plot tdqvec tdcvec tcqvec vs tdvvec
let min_tdq = tdqvec[0]
let min_index = 0
let index = 1
while index<Nsim
    if tdqvec[index] < min_tdq
        let min_tdq = tdqvec[index]
        let min_index = index
    end
    let index = index+1
end
let min_tdc = tdcvec[min_index]
let min_tcq = tcqvec[min_index]
let min_tdv = tdvvec[min_index]

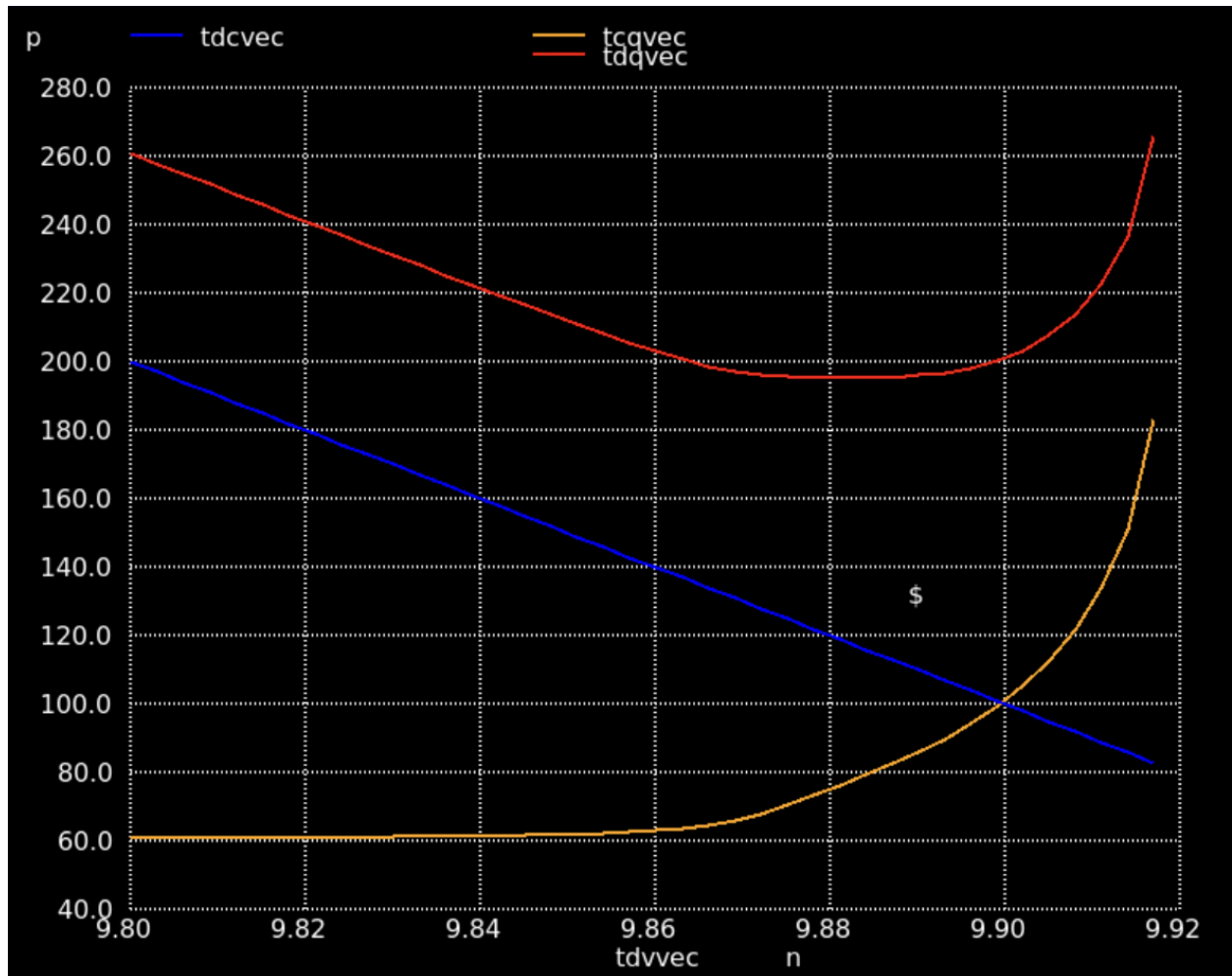
echo min_tdq : $&min_tdq min_tdc : $&min_tdc min_tcq : $&min_tcq min_tdv : $&min_tdv

.endc

```

Code used to calculate T-setup Clock-q delay at rising edge

Plots :-



- The time at which tDQ is minimum was found to be $9.881E-09s$.
- The tDQ was found to be $1.95334E-10s$.
- The tCQ was found to be $7.63337E-11s$.
- The tDC was found to be $1.19E-10s$.

Question 1.B(Falling edge):

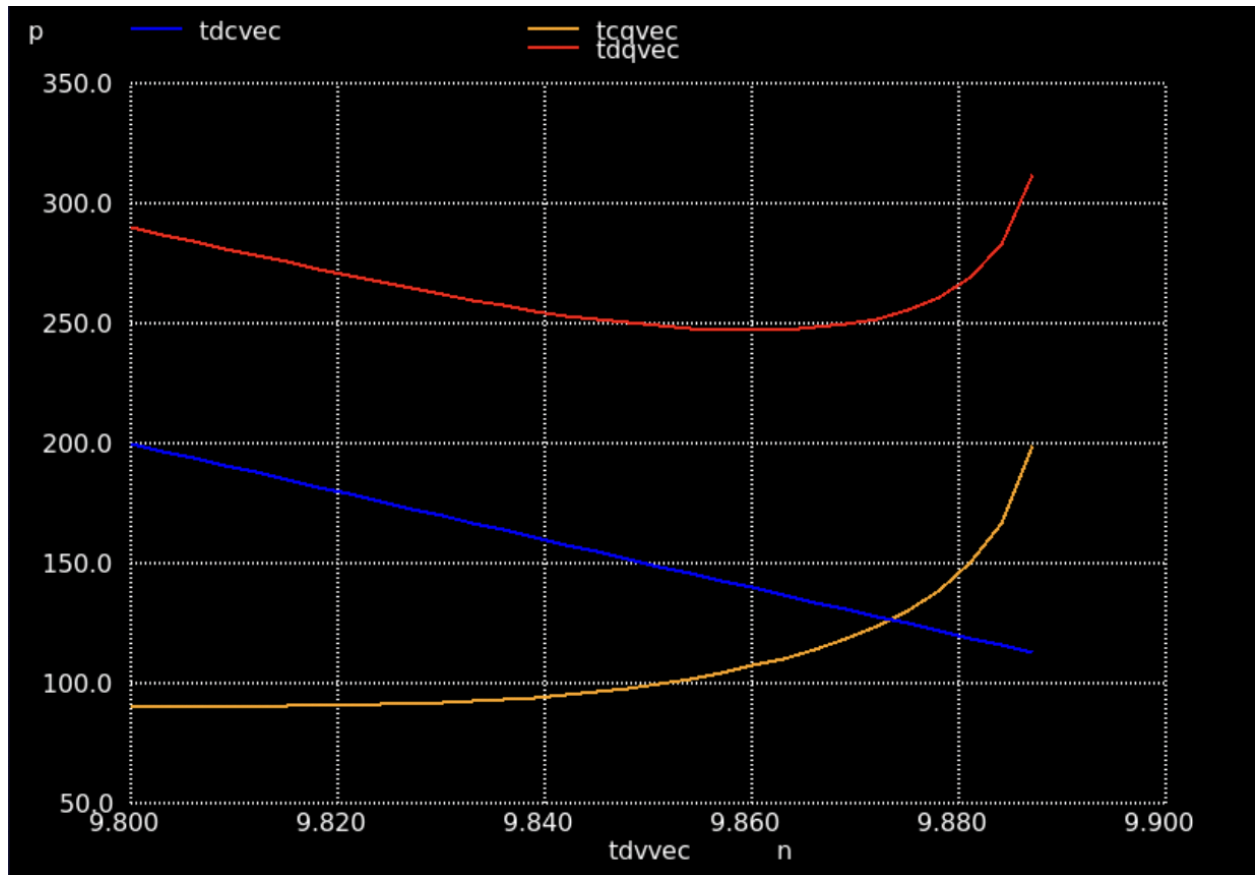
```
s1
.param td = 0.5n
.ic V(Vout) = 1.8
.ic V(Vd) = 1.8
.control
let Nsim = 30
let tdcvec = vector(Nsim)
let tcqvec = vector(Nsim)
let tdqvec = vector(Nsim)
let tdvvec = vector(Nsim)
let index = 0
while index<Nsim
    let tdv = 9.8n + (index*((0.12n)/40))
    alterparam td = $tdv
    reset
    tran 2p 50n
    meas tran tdc trig v(Vd) val=0.9 fall=1 targ v(phi) val=0.9 rise=1
    meas tran tcq trig v(phi) val=0.9 rise=1 targ v(Vout) val=0.9 fall=1
    let tdcvec[index] = $tdc
    let tcqvec[index] = $tcq
    let tdqvec[index] = $tdc + $tcq
    let tdvvec[index] = tdv
    let index = index+1
end
plot tdqvec tdcvec tcqvec vs tdvvec
let min_tdq = tdqvec[0]
let min_index = 0
let index = 1
while index<Nsim
    if tdqvec[index] < min_tdq
        let min_tdq = tdqvec[index]
        let min_index = index
    end
    let index = index+1
end
let min_tdc = tdcvec[min_index]
let min_tcq = tcqvec[min_index]
let min_tdv = tdvvec[min_index]

echo min_tdq : $min_tdq min_tdc : $min_tdc min_tcq : $min_tcq min_tdv : $min_tdv

.endc
```

Code used to calculate T-setup Clock-q delay at falling edge

Plots :-



- The time at which tDQ is minimum was found to be $9.86E-09s$.
- The tDQ was found to be $2.47522E-10s$.
- The tCQ was found to be $1.07522E-10s$.
- The tDC was found to be $1.4E-10s$.